RSOC-2026

Group-30

- ProblemStatementID:– **04**
- ProblemStatement Title:- **CollabCode: Real-Time Collaborative Coding Platform**
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- TeamID:- **30**
- TeamName:- **Team 07**
- DomainName:- **Web Development**

IDEA TITLE

Proposed Solution: CollabCode

The Real-Time IDE for Seamless Peer Programming

1. Detailed Explanation

- **Cloud-Synced IDE:** A unified multi-user workspace for real-time coding, debugging, and execution.
- **Built-in Collaboration:** Features independent cursors, shared terminals, and integrated voice/video.

2. How It Addresses the Problem

- **Zero Context Switching:** Merges dev work and communication into one platform.
- **Eliminates "Merge Hell":** Prevent version conflicts with a single, live-synchronized codebase.
- **Direct Mentoring:** Enables instant "follow-along" coding for interviews and peer training.

3. Innovation and Uniqueness

- **Collective AI Pilot:** A shared assistant aware of the entire team’s real-time context.
- **Follow-Me Mode:** Viewport anchoring to the lead developer’s cursor with zero latency.
- **Instant Sandboxing:** One-click deployment to a live URL for immediate collaborative tes

FEASIBILITY AND VIABILITY

1. Analysis of Feasibility

- **Proven Tech Stack:** Uses **WebSockets** and **CRDTs** (Conflict-free Replicated Data Types) for real-time, lag-free synchronization.
- **Cloud Infrastructure:** Scalable via AWS/Azure to ensure high availability and global access.

2. Potential Challenges and Risks

- **Latency:** Network lag affecting real-time cursor tracking.
- **Concurrency:** Handling simultaneous file edits or terminal commands.
- **Security:** Potential risks in executing shared code in a common environment.

3. Strategies for Overcoming Challenges

- **Edge Computing:** Deploying regional nodes to minimize latency.
- **Docker Sandboxing:** Isolating each session in a secure container to prevent system-wide breaches.
- **OT Algorithms:** Implementing robust Operational Transformation to resolve edit conflicts

IMPACT AND BENEFITS

1. Potential Impact on Target Audience

- **Developers:** Boosts productivity by enabling real-time collaboration and eliminating "merge hell."
- **Students/Learners:** Accelerates learning through direct, live shadowing of experienced mentors.
- **Tech Recruiters:** Streamlines technical interviews with a native, shared coding environment.

2. Benefits of the Solution

- **Social:** Democratizes expert knowledge by making remote peer-mentorship accessible and interactive.
- **Economic:** Reduces operational costs for companies by shortening development cycles and onboarding time.
- **Productivity:** Consolidates communication and coding into one hub, removing friction and context switching.

RESEARCH AND REFERENCES

1. Core Technical Research

Sync Logic: Utilized **OT (Ellis & Gibbs, 1989)** and **CRDTs** for conflict-free, real-time code editing.
Latency: Optimized via **Edge Computing** and **WebSockets** for sub-100ms response times.
Security: Implemented **Docker-based sandboxing** for isolated, secure code execution.

2. Industry Benchmarks

UX Design: Analyzed **VS Code Live Share** and **Google Docs** for seamless multi-cursor tracking.
AI Pair-Programming: Integrated **LLM research** to power the shared, context-aware AI assistant.
Educational Impact: Based on **Distributed Pair Programming** studies showing 40% higher knowledge retention.